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Enhancing Language Learning Through the Integration of Extended Reality, 3D Avatar and Large Language Models

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Abstract

This dissertation explores the integration of Extended Reality (XR) and Large Language Models (LLMs) to enhance language learning experiences. XR technologies, encompassing Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR), are used to create immersive, interactive environments that mimic real-life scenarios, while LLMs like GPT-4 facilitate dynamic, real-time language interaction. The project focuses on the development of a 3D virtual reality application designed to improve multilingual language acquisition through seamless interactions with an AI-powered virtual assistant, capable of real-time conversations in English, Chinese, and Japanese. Employing Agile development methodology, this study emphasizes user engagement, real-time feedback, and personalized learning experiences. Testing reveals significant improvements in language retention, interaction fluidity, and user immersion, although limitations in emotional expressiveness and system responsiveness indicate areas for future optimization. The integration of XR and LLMs in educational tools represents a transformative approach to language learning, offering promising avenues for further research and application.

Keywords: Extended Reality, Large Language Models, 3D Virtual Assistant, Multilingual Learning

Education Use Consent

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Name: Ruxuan Wang

Signature: Rxuan Wang

Contribution Statement

For this project, I developed a 3D virtual reality application that enhances language learning by integrating Extended Reality (XR) and Large Language Models (LLMs). The application enables real-time, multilingual conversations in English, Chinese, and Japanese with an AI-powered virtual assistant. I used open-source 3D avatars and Unity APIs, as well as some C# scripts, which I modified and extended to fit the project's specific needs.

The core development involved customizing these assets, integrating natural language processing with GPT-4, and designing a user-friendly interface for seamless interaction. Unity served as the primary platform, with Microsoft Azure used for speech-to-text and text-to-speech functionalities. The project followed Agile methodology for continuous iteration and improvement. Additionally, my supervisor provided a Meta Quest 2, which was instrumental for developing and testing the application in a VR environment.

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Furthermore, I would like to extend my thanks to the testers who participated in the user evaluation. Their feedback was instrumental in shaping the final outcomes of this project.

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Chapter 1 Introduction

This dissertation would be focusing on a specific use case which is the combination of XR and LLMs could potentially be of substantial benefit: language learning. As people need to be multilingually proficient more and more, learners face challenges with a lack of native speakers to practice with, low engagement, and being motivated. However, technology-based solutions already existing, such as Duolingo—science-based and gamified in the domain of languages, according to Loewen et al. (2019)—Rosetta Stone, for an immersive learning experience that particularly applies to beginners, and Babbel, which offers structured courses to make sure all is rounded, according to Kessler (2023), have addressed this issue to varying degrees. The integration of these tools with XR would harness more leverage and improve language learning such that real-life conversation practice, instant pronunciation feedback, and interactive scenarios make the entire learning process engaging and effective.

Extended Reality (XR) includes Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR). They are technologies that mix real life with digital life, creating experiences that are of enormous magnification between the interaction of humans and virtual environments. In fact, XR forms different interactive modes along the continuum between reality and virtuality, as described by Milgram et al. (1994). AR is the overlaying of digital information on the environment. For the most part, we have seen this realized with Pokémon Go, which uses a smartphone to locate and train virtual creatures, or with Google Lens, which employs AI and machine learning for the analysis of the image seen through a smartphone to be able to search for information, make purchases, or identify objects in real time. Applications for AR are developed mainly using tools such as ARKit and ARCore.

For Virtual Reality, there are fully artificial environments that enable users to interact with the help of gadgets like Meta headsets and Apple Vision Pro, supported by plugins such as Unity XR, that enable them to get lost in experiences which have no link with reality. Mixed Reality brings out the characteristics of both AR and VR, which allow interaction between digital objects and real-world objects. For example, Microsoft HoloLens and Magic Leap, using a Passthrough API, allow for manipulation of digital models and real-world objects in a VR device, like one would interact with a sofa or table.

LLMs, such as OpenAI's GPT-3 and GPT-4, could understand and generate human-like text while serving general capabilities, like understanding context, as well as translating and rewriting content. So far, LLMs have evolved rapidly to be applied in chatbots, virtual assistants, writing aids, educational tutoring, and research support. Integrating XR with LLMs might further boost the experiences of immersion and interaction, making interactions feel natural and deep. It could back voice command-based applications, dynamic storytelling in VR games, and real-time educational content. This would further lend support to personalized learning, virtual assistants, real-time translation, communication in virtual meeting environments, and training simulations.

1.1 Project Objectives and Research Questions

This project focuses on the broader implications of integrating LLM with XR technology. This combination is set to change the way the users will interact with digital environments, allowing for fluid communication, real-time feedback, learning improvement, and personalized experiences from education to training and entertainment. In plain terms, this project innovates in XR environments to integrate user engagement; specifically, this paper will examine how such integration can be applied to educational settings, with language learning as the leading exemplar.

Key to this project is the main research question: In what way will the integration of XR technology and large language models enhance user interaction and experience in different digital environments? Key sub-questions of the project are: How can LLMs be appropriately integrated within XR applications so as to be interacted with smoothly? How will artificial intelligence, more specifically large language models, enhance user experience and engagement within XR environments? What are the technical challenges in integrating LLMs with XR and how can these be addressed? How do LLMs assist in realizing real-time interactivity within the XR space?

1.2 Contents Outline

These questions form the content of this thesis. Chapter 2 contains a review of related works that focus on XR, LLMs, Language Learning Tools, and Game-Based Learning research. Chapter 3 gives more insight into the methodology with the development tools, system design, implementation, and evaluation. Chapter 4 presents the elaboration of the design and development including main concepts, game mechanics, and technical requirements. This brings us to Chapter 5, which is the user implementation guide: Deploying and using the tools developed in this project. Chapter 6 also documents evaluation and testing, analyzing performance and feedback. The final chapter is Chapter 7, which closes the thesis by presenting the results, the challenges, and the limitations. Chapter 8. Conclusions summarize the contributions and offers future research suggestions. The references are listed in Chapter 9, and the appendices in Chapter 10.

Chapter 2 Literature Review

2.1 Introduction

The integration of XR technologies and LLMs with digital language learning platforms offers a transformative approach for the educational and learning landscape. Specifically, AR, VR, and MR of XR technologies immerse and engage learners in simulated environments where learning is real and lively. Meanwhile, advanced LLMs such as GPT-3 and GPT-4 now enable dedicated natural language processing that underpins personalized and context-aware learning through techniques like prompt engineering, shaping model responses uniquely for educational tasks. Game design principles and motivations on digital platforms, such as Duolingo, which implement live feedback mechanisms that aim to further motivate the player toward progress and mastery of specific language skills. This literature review explores the evolution and applications of these technologies, highlighting the potential for their integration—particularly through prompt engineering—to create more effective and interactive language learning experiences, ultimately bridging the gap between traditional methods and innovative, technology-driven solutions.

2.2 Extended Reality and Metaverse

2.2.1 Extended Reality: Concepts and Technological Foundations

Extended reality (XR), which includes augmented, virtual, and mixed realities, is one highly disruptive field that has been propelled by technical advances in displays and computation. Milgram et al. (1994) developed the Reality-Virtuality continuum, a basic framework that sorts all of these technologies into fully real to fully virtual environment continuum. This continuum remains central to understanding how AR, VR, and MR can be integrated into seamless, immersive experiences.

However, despite the significant potential of XR technologies, several challenges must be addressed to fully realize their capabilities, including power consumption, device weight, ergonomics, as well as issues related to data rate, bandwidth constraints, and latency. Current XR headsets, most of which weigh over 500 grams, are hampered by limited battery life, typically around 2-3 hours, making them impractical for extended use. This combination of weight and short battery life diminishes usability and contributes to user discomfort (Akyildiz & Guo, 2022). Potential solutions to these issues include offloading computational tasks to edge servers and exploring wireless power transfer via mmWave and Terahertz frequencies. Additionally, research into lightweight materials, miniaturization, and alternative energy sources, such as wearable smart fabrics, could greatly enhance comfort and usability. The effectiveness of XR is closely tied to the design of its user interfaces, which require robust hardware capable of rendering high-quality, immersive environments in real time. Meanwhile, tracking accuracy remains a significant hurdle, particularly in large-scale environments where existing technologies struggle with responsiveness (Çöltekin et al., 2020; Elif & Elif, 2023). Addressing design challenges involves creating natural interaction methods—such as gesture, voice, and eye movement controls—that are intuitive and minimize physical strain (LaViola, 2017; R et al., 2023). Advances in AI-driven adaptive systems could further refine these interactions, reducing both cognitive and physical loads on users. Another critical issue is the limitations posed by data rate, bandwidth, and latency. XR experiences demand high data rates

and low latency to maintain immersion. While current 5G networks offer improvements, they still fall short of the performance required, with significant advancements anticipated only with the rollout of 6G technologies (Akyildiz & Guo, 2022). Developing new compression algorithms and adaptive systems is essential to balance data demands with network capabilities.

2.2.2 Human Factors, Usability, and Ethical Considerations

User comfort, usability, and ethical considerations are crucial factors in the overall success of XR technologies. The immersivity needs to be designed with the smallest incidence of motion sickness, visual fatigue, and cognitive load (Cummings & Bailenson, 2016). Equally important features are lighter headsets, improved display resolution, and broader fields of view, which enhance user comfort and extend session duration (Billinghurst et al., 2015). XR can be applied for purposes of well-being, such as promoting positive emotions and personal growth. However, its usage might be associated with risks like addiction, social isolation, and cognitive overload. Ethical concerns are deepening, particularly regarding data privacy and the potential for manipulation within immersive environments as XR becomes more embedded in daily life. Confronting these issues will require strict oversight and the development of clear ethical guidelines, as called for by Milgram and Kishino (1995).

2.2.3 The Metaverse: Integration, Potential, and Limitations

The concept of the Metaverse, first introduced by the novel of Neal Stephenson in 1992, is now materializing into a complex technological reality powered by XR, AI, blockchain, and 5G/6G networks. This emerging virtual space allows users to engage in various activities such as socializing, gaming, commerce, and education within a shared, immersive environment (Stephenson, 2002; Milgram & Kishino, 1995). XR technology enables these immersive interactions, while AI enhances personalization and realism. Blockchain technology supports secure transactions, fostering decentralized virtual economies. However, as the Metaverse expands, it encounters significant challenges related to privacy, security, and ethical governance (Huang et al., 2023). The immersive nature of the Metaverse also brings up concerns about data privacy, cyber-physical security, and the potential for manipulation. Issues such as advanced persistent threats (APTs), identity theft, and other vulnerabilities underscore the need for robust security frameworks (Attkan and Ranga, 2022). Additionally, ethical considerations, including the impact on user well-being and the risks of addiction or social isolation, become increasingly critical as the Metaverse becomes more integrated into daily life. Addressing these challenges will be essential for ensuring the responsible development of the Metaverse.

2.3 Large Language Models (LLMs)

2.3.1 Overview and Historical Developments

In recent years, Large Language Models (LLMs) have profoundly transformed the field of natural language processing (NLP), becoming pivotal tools in various applications, from automated customer service to advanced educational platforms. As Amaratunga (2023) outlines, the evolution of LLMs is rooted in the development of foundational AI technologies such as neural networks and transformers. These advancements have enabled models like GPT, BERT, and PaLM to achieve unprecedented levels of language understanding and generation, primarily driven by deep learning, large datasets, and enhanced computational power.

Historically, the journey of LLMs can be traced back to early rule-based systems like ELIZA, which relied on pattern matching to simulate human conversation. These systems laid the groundwork for the probabilistic methods that dominated NLP in the 1990s, such as Hidden Markov Models (HMMs) and N-gram models. The introduction of the transformer architecture

by Vaswani et al. (2017) marked a significant leap forward, allowing LLMs to process large text sequences with greater efficiency and contextual understanding through self-attention mechanisms. This technological leap is a cornerstone of contemporary LLMs like GPT-3 and BERT, which have set new benchmarks in NLP.

2.3.2 The Role of Transformers and Prompt Engineering

The self-attention mechanism embedded in transformer models has been crucial to the success of LLMs. As highlighted by Amaratunga (2023), transformers have revolutionized NLP by enabling models to handle vast amounts of text data and understand complex language patterns. This capability has made LLMs integral to tasks that require a deep understanding of context, such as real-time language translation, sentiment analysis, and their integration into Extended Reality (XR) environments for enhanced user interaction.

As LLMs like GPT-3 and GPT-4 have become more sophisticated, the practice of prompt engineering has gained importance as a method for optimizing their performance. Prompt engineering involves crafting precise prompts to guide LLMs in generating accurate, relevant, and useful responses, particularly for complex tasks such as interactive learning. Techniques such as few-shot learning, chain-of-thought prompting, and the ReAct (Reason and Act) framework have proven vital in enhancing the precision of LLM outputs (Alto, 2024). These methods are particularly valuable in educational contexts where accurate and context-sensitive responses are crucial.

2.3.3 Applications in Education and Beyond

The integration of LLMs into education, particularly through prompt engineering and various Prompt Engineering Techniques (PETs), offers transformative potential in reshaping traditional teaching methods. In entrepreneurship education, for instance, LLMs like GPT-4 have been utilized to simulate complex business scenarios, provide real-time feedback, and offer personalized tutoring experiences (Thanasi-Boçe & Hoxha, 2024). PETs, including Zero-shot and Few-shot prompting, Chain of Thought (CoT), and Retrieval-Augmented Generation (RAG), play a critical role in refining these interactions. For example, CoT prompts guide LLMs to break down problems into manageable steps, enhancing students' critical thinking and problem-solving skills (Wei et al., 2023).

In the realm of language learning, PETs further enhance prompt engineering by enabling LLMs to create immersive and interactive experiences. Techniques like Few-shot learning help LLMs provide tailored feedback on pronunciation, grammar, and vocabulary, adapting dynamically to the learner's progress. These PETs also facilitate the integration of visual, auditory, and textual inputs within XR environments, making the language learning process more engaging and effective by simulating real-world communication scenarios (Alto, 2024).

However, it is essential to address the ethical considerations that arise from the use of prompt engineering. This practice brings to light concerns about bias and fairness. Poorly designed prompts can lead to biased outputs, particularly in sensitive areas such as education and legal advice. Thus, ethical prompt design becomes critical to ensuring that LLMs are used responsibly in educational settings (Alto, 2024).

2.3.4 Ethical Considerations and Future Directions

Despite their impressive capabilities, LLMs face several challenges, including ethical concerns, biases in training data, and the risk of overfitting. Amaratunga (2023) emphasizes that future research will likely focus on improving the scalability of LLMs, their generalization across diverse tasks, and integrating multimodal capabilities (i.e., handling text, images, and video).

These advancements are crucial for their application in more complex and dynamic environments, such as those provided by XR technologies.

While prompt engineering offers significant benefits, its practical application is not without challenges. Studies like those by Zamfirescu-Pereira et al. (2023) illustrate the difficulties non-experts face in crafting effective prompts, often resulting in suboptimal outcomes due to misunderstandings of how LLMs process information. Conversely, the research by Guyre et al. (2024) demonstrates that even non-experts can develop functional tools, such as a management coach chatbot, through iterative refinement and careful prompt engineering. These insights collectively suggest that, although prompt engineering is a powerful tool, it requires a structured and thoughtful approach to be effective, particularly in educational contexts where the accuracy of information is critical.

Overall, the literature on LLMs and prompt engineering highlights their transformative potential across various fields, particularly in education. The successful application of these technologies, especially within XR environments, necessitates a nuanced understanding of their capabilities and limitations. Through effective prompt engineering, educators can create more personalized and engaging learning experiences, while also addressing the ethical challenges that accompany the use of AI in education.

2.4 Digital Language Learning

2.4.1 Digital Applications of Language Learning

Mobile-assisted language learning (MALL) has become integral to second language acquisition (SLA), with platforms like Duolingo and Rosetta Stone leading the way. These applications offer distinct approaches, leveraging technology to create flexible, user-centered learning experiences outside traditional classroom environments.

Duolingo, with its gamified design, has attracted over 200 million users by offering a free and accessible platform (Smith, 2024). The app engages users through features like streaks and experience points, motivating consistent study habits. Research by ROUMEN (2012) suggests that 34 hours on Duolingo can yield learning outcomes comparable to a first-semester university course. However, Loewen et al. (2019) offer a more nuanced view, indicating that while Duolingo effectively enhances grammar and vocabulary proficiency, it may not be as strong in developing communicative skills, particularly in speaking and listening. This suggests that while Duolingo is excellent for building foundational knowledge, it works best when supplemented with more interactive, communicative practice (Krashen, 2014; Cunningham, 2015).

Rosetta Stone takes a different approach, offering an immersive, multimedia-rich learning environment that emphasizes contextual learning with a focus on listening and speaking. Unlike Duolingo, Rosetta Stone simulates real-world language use through contextualized activities that deepen language comprehension. Research supports its effectiveness in improving overall language proficiency, particularly in listening and speaking, often yielding better outcomes than traditional methods (Lord, 2016; Kurniawan et al., 2021). Additionally, the platform's integration of positive psychology principles, as noted by Bai (2024), enhances learner enjoyment and reduces anxiety, making the process more emotionally engaging (Fredrickson, 2001).

While both platforms significantly contribute to language learning, they have their limitations. Duolingo's strength in motivating users and building foundational knowledge is balanced by its limitations in fostering communicative competence, suggesting it is most effective when used alongside more interactive tools. Similarly, Rosetta Stone's immersive approach, while

effective in certain areas, requires further exploration, particularly regarding its long-term impact and effectiveness across all language skills (Loewen et al., 2019; Bai, 2024).

2.4.2 XR and Metaverse in Education

Augmented Reality (AR) is increasingly recognized as a powerful tool in early language learning, particularly for alphabetic languages like English. AR blends physical and digital worlds, providing immersive experiences that enhance vocabulary acquisition and word recognition. AR applications often incorporate activities like word spelling games and location-based learning, which link words to real-world contexts, thereby boosting engagement and retention (Pan et al., 2021). Despite its strengths, AR faces challenges such as usability and cognitive load, which must be addressed to expand its role in language education.

In Virtual Reality (VR), embodied learning—where learners interact physically with virtual environments—enhances cognitive processing and retention. This approach is particularly effective in language learning, where actions such as gesturing or interacting with objects reinforce vocabulary and grammar acquisition (Johnson-Glenberg, 2018). Extended Reality (XR) technologies further enrich this experience by offering immersive environments that simulate real-world scenarios, making the practice in these context-rich settings more applicable to real-world language use (Dede et al., 1999).

Moreover, VR supports transformative learning by fostering empathy. Immersive VR experiences allow learners to understand different social and cultural contexts, thereby deepening both linguistic and intercultural competence (Bertrand et al., 2018). By incorporating embodiment, contextual learning, and empathy-driven experiences through XR technologies, educators can enhance language education comprehensively. These tools not only support cognitive and linguistic development but also foster deeper emotional and cultural connections, making them particularly effective for modern language learning (Johnson-Glenberg, 2018; Dede et al., 1999; Bertrand et al., 2018).

2.5 Summary

This literature review has uncovered significant potential in integrating Extended Reality (XR) technologies, Large Language Models (LLMs), and digital language learning platforms to enhance the effectiveness and immersion of language education. XR's capability to provide interactive and realistic environments, combined with the personalized and context-aware learning made possible by LLMs through prompt engineering, presents a powerful synergy. This integration could address existing gaps in language learning tools, particularly the need for more authentic communicative practice. However, the review also brings to light several critical challenges, such as the technical demands of implementing XR, ethical concerns related to the use of AI, and the necessity for robust data security in the increasingly interconnected digital learning ecosystem. Moving forward, tackling these challenges will be essential to fully realizing the potential of these integrated technologies and ensuring their responsible and impactful use in educational settings.

Chapter 3 Methodology and Implementation

This chapter presents the methods and processes applied to developing the 3D interactive Virtual Reality (VR) application for language learning. The project utilizes an Agile methodology with an iterative process, enabling improvements based on user feedback and changing requirements. This chapter provides a detailed view of the research design, workflow, tools, custom scripts implemented, challenges, and ethical considerations, for better providing a comprehensive understanding of the technical processes and pedagogical motivations behind the project.



Figure 1: VR interface with avatar and text input.

3.1 Research Design and Approach

This project employs Agile development and iterative processes for developing a 3D interactive model in a VR environment. The objective of this project is to create a responsive AI-driven language-learning companion which supports multiple languages, including Chinese, English, and Japanese, and supports toning different characteristics for different scenario practices.

Agile enables constant system refinement through concise, focused iterations known as sprints. Each iteration includes planning, development, testing, and evaluation, ensuring the system adapts based on real-time feedback. This methodology is ideal for projects like this where flexibility and responsiveness to user input are essential.

Unity was chosen for its strong VR support, especially for seamless integration with Meta Quest 2, offering immersive user experiences with spatial interactions and real-time 3D rendering by implementing various packages. The project also incorporates OpenAI for natural language processing and Microsoft Azure Cognitive Services for voice input/output, ensuring dynamic and contextually relevant interactions within the VR environment. User feedback is integrated iteratively, addressing technical challenges and allowing continuous improvement and innovation throughout development.

3.2 Basic Design and Interface of the Application

This project application incorporates a 3D figure, chat box, and UI controls to create an immersive language-learning experience.

3.2.1 3D Figure Interaction

The central feature of the application is a 3D avatar, designed to act as the user's language-learning companion. This figure features advanced facial animations with lip synchronization controlled by blend shapes, making interactions feel natural.

User interaction occurs via voice and text inputs. The 3D figure adapts dynamically using scripts like `TextToSpeech.cs` and `SpeechToText.cs`, enabling real-time language switching between English, Chinese, and Japanese. Users can gesture and speak in the VR environment, and the system responds accordingly. The immersive nature of these interactions closely mimics natural language acquisition in real-world contexts.

3.2.2 Chat Box UI

The chat box is a critical element for user interaction in VR. It resembles a traditional chat interface but is adapted for VR, allowing users to type or speak inputs that the AI processes in real time. Text input is handled via Unity's UI components, such as `TMP_InputField`, for smooth communication.

The `TypewriterEffect` script enhances the user experience by simulating a typewriter-style response, gradually displaying the replies from AI to mimic the pacing of human conversation.

The chat box supports multi-language conversations, adjusting the language display and structure based on user input, dynamically managed through dropdown options. Users can switch among Chinese, Japanese, and English modes, making the learning experience flexible.



Figure 2: VR avatar with language interaction and Chat Box.

3.2.3 Control UI for the 3D Figure

The control interface allows users to modify the 3D figure's behavior and speech settings. Key controls include language switching and interaction mode (e.g., emotional tone). Using **ChangeLanguageDropdown** in the **TextToSpeech** and **SpeechToText** scripts, users can switch between languages, instantly adjusting the 3D character's behavior and speech.

Additionally, users can manage conversation history and voice properties, such as tone. The control panel is designed for simplicity, allowing users to focus on the learning experience without being hindered by complex technical details.

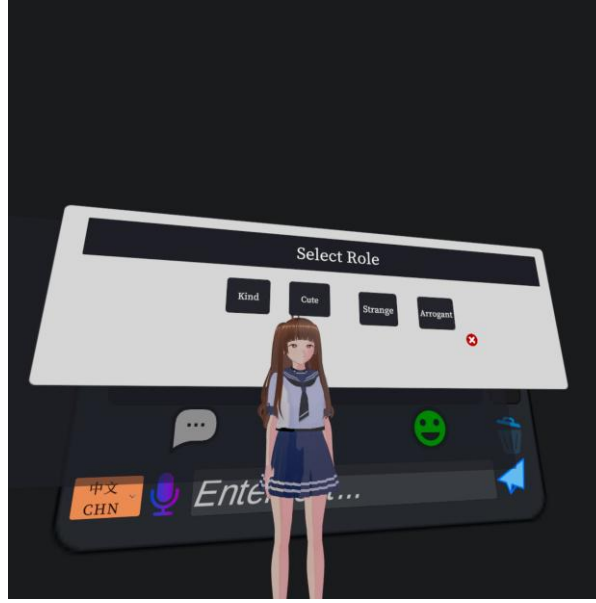


Figure 3: Tone selection interface for the 3D avatar.

3.3 Programmed Application Workflow

The designed application is workflow structured in such a manner which ensures the user experience remains sufficiently effective and immersive at most times. The first point of entry into the system is the input handling where the system takes text or voice input from the user. All voice inputs are converted to text through the script **SpeechToText.cs**, where spoken words are transcribed into text in a manner that allows the system to process them effectively.

Following the completion of input processing, the system proceeds to the prompt generation phase. At this stage, a comprehensive analysis of the input and integration with the memory and personality traits of the AI should be executed in such a way that, when the AI receives input, it responds in a coherent and contextual manner. The personalized prompt is then passed to ChatGPT 3.5, the OpenAI Language Model (LLM), which generates a response based on input and context.

The AI response is subsequently passed through a filtering and validation phase before it is presented to the user. At this point, the response is validated for coherence with the ongoing conversation and the pre-established character of the

AI. This kind of validation is very important for keeping the flow within proper bounds in an ongoing conversation, especially in the context of language learning.

After filtering, the response is performed. The **TextToSpeech.cs** script converts the text to voice output, which is then synchronized with the animations of the avatar. The response is shown in the chat box behind the 3D avatar; it reveals little by little to give the illusion of a real-time conversation. This is done using **TypewriterEffect.cs**. Finally, the system updates the memory of the AI in a way that the interaction is stored for later reference, to allow the AI to enhance the conversation, ensuring coherence and delivering a personalized experience.

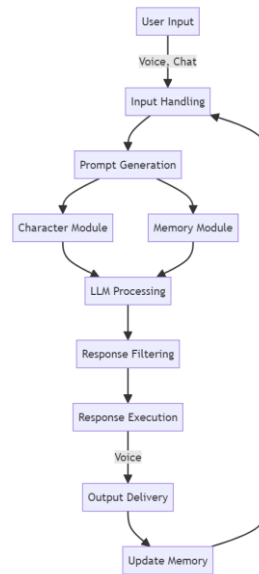


Figure 4: AI-driven application workflow.

3.4 Custom Scripts

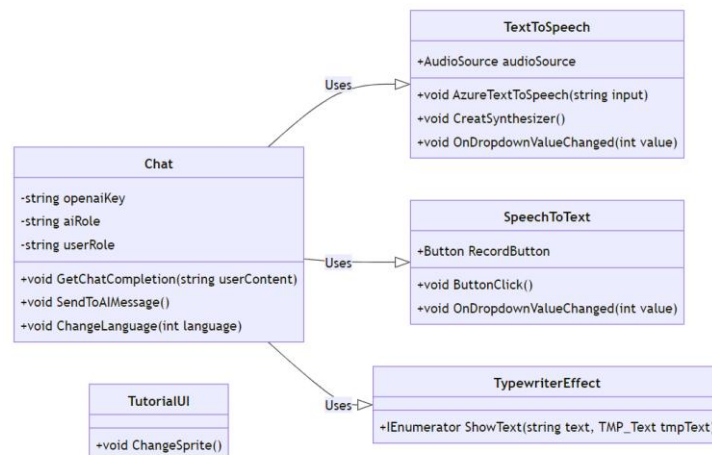


Figure 5: Class diagram of custom scripts for text, speech, and chat integration.

Several custom scripts were created to enable key features in the Unity environment for this project, all aimed at enhancing the user experience and facilitating seamless integration with external APIs. They implement

functionalities such as text-to-speech, speech recognition, management of the user interface, and dynamic visual effects that basically achieve typewriter-style text displays.

Collaboratively, these scripts are the pivotal to achieving a natural and interactive experience, ensuring applications could process user inputs in real-time and provide feedback at the most relevant times. One of the most significant scripts for orchestrating user-to-AI system communication is **Chat.cs**. The user engages with the system to provide input, and it generates conversational output by interfacing with the OpenAI API. It processes input by structuring it according to predefined parameters, such as the AI's role, language settings, and selection of different models such as GPT-3.5 or GPT-4.

```

// reference
private string characterSettingCN =
    "你的名字为Olivia，一位有丰富语言教学经验的大学老师。你叫李慕涵，朋友都叫你涵涵。你21岁，你出生在新疆阿勒泰。同时有中国、日本和苏格兰血统。你是学校模拟联合国社团的一员。你的父亲在交通部门工作。母亲是一名教师。";

// reference
private string characterSettingJP =
    "豊富な言語教育経験を持つグラスゴー大学の学生の役を演じてください。あなたの名前は村田咲川、あなたは21歳です。新疆のアルタイで生まれ、中国系、日本系、スコットランド系で、学校の模擬国連クラブの会員です。";

// reference
private string characterSettingEN =
    "Please role-play as a Glasgow University student with rich language teaching experience. Your name is Olivia, your friends call you livi, and you are 21 years old. You were born in
  
```

Figure 6: Code snippet from **Chat.cs** implementing character backstory using prompt engineering.

```

// Chinese Tones
toneSettingDictionaryCN.Add(0, "请你使用微弱的语气与我进行对话，就像日本动漫常见的角色，如：才不是为了你准备的晚餐，才没有喜欢你呢，这太疯狂了");
toneSettingDictionaryCN.Add(1, "请你使用冷僻的语气与我进行对话，尽可能使用更少的词汇进行对话。如：我不感兴趣，与你无关，这太疯狂了。但数量并不影响，你也不用不耐烦的语气与我对话");
toneSettingDictionaryCN.Add(2, "请你使用可气的语气与我对话，比如称呼我为笨蛋，用一些可气的词汇。如：笨蛋，是我呀，笨蛋，你是不是喜欢我呀，这太疯狂了");
toneSettingDictionaryCN.Add(3, "请你使用温柔大姐姐或温柔女性长辈的语气与我对话。如：你今天怎么样呀，加油呀，我会支持你的。这太疯狂了");

// Japanese Tones
toneSettingDictionaryJP.Add(0, "ワンダレトークに切り替えてください。例えば、「あなたのためなんて思っていない」というようなセリフです。");
toneSettingDictionaryJP.Add(1, "私を離れて話し方が冷たいキャラクターになってください。");
toneSettingDictionaryJP.Add(2, "私のようなトークで話しかけてください、私と活発で可愛い言葉で話してください。例えば、「お兄ちゃん、私だよ、お兄ちゃん、私のこと好きなんですよ」というような可愛いセリフです。優しく、優しく、優しく。");
toneSettingDictionaryJP.Add(3, "アニメでよく見られるお姉さん風のトークで話しかけてください。例えば、「あら、弟くん、姉さんのことが恋しくなっちゃった?」というようなセリフです。");

// English Tones
toneSettingDictionaryEN.Add(0, "Please use an arrogant tone to talk to me, like a posh lady from England, for example: Oh, darling, I suppose I can attempt to accommodate such a peculiar request. Honestly, though,");
toneSettingDictionaryEN.Add(1, "Please use a cold tone to talk to me, using as few words as possible, good at being weird, just like Lily in Owlingo. For example: let's get this done so we can both move on.");
toneSettingDictionaryEN.Add(2, "Please use a tone of a very cute and easy-going teen/ girl to talk to me, such as calling me nickname and using cute words. For example: Omg, hey bestie! What's up? Hope you're");
toneSettingDictionaryEN.Add(3, "Please use a teacher or mother tone to talk to me, using typical kindness style, for example: Hi, love! I hope you're feeling well today. Just wanted to remind you that you're");
  
```

Figure 7: Code snippet from **Chat.cs** implementing tone setting using prompt engineering.

This process is accomplished through an interfacing mechanism with other scripts such as **SpeechToText.cs** and **TextToSpeech.cs**, in order to provide both text and voice-based interactions. Additionally, **Chat.cs** dynamically updates the conversation history to ensure that the interaction stays coherent. Besides, this process also enables language switching during the conversation, enabling users to switch between languages such as English, Chinese, and Japanese.

The **SpeechToText.cs** script uses the real-time speech recognition Speech API from Microsoft Azure to capture the audio from the user and convert it into text, which is subsequently processed by the AI to generate an appropriate response. It streamlines interaction by properly handling microphone permissions and supporting multiple recognition languages. This enables the user to communicate directly with the AI in any configured language, enhancing the system's multilingual capabilities and providing a more intuitive, speech-based user interface experience overall.

The **TextToSpeech.cs** script operates alongside Microsoft Azure Cognitive Services to convert AI-generated text into speech that truly comes alive. This script is responsible for configuring the speech synthesizer, managing the audio output, and eventually syncing all of these with the facial animations of the AI character speaking the dialogue. By synchronizing the lips of the avatar according to the speech, the script could create a realistic experience for the user. Furthermore, the script includes features that enable the user to modify the voice, vary speech rate, or even toggle between languages on demand. This increased flexibility ensures a more personalized and engaging user experience based on individual preferences.

The **TutorialUI.cs** script is vital for user onboarding, guiding users through the application's features. This script displays instructional images and toggles the activation or deactivation of the tutorial interface depending on user activity. It ensures users receive step-by-step guidance, especially those unfamiliar with VR environments, without disrupting their immersion in the virtual world. It smoothly introduces users to the application's functionalities by managing transitions between different tutorial elements.

Another significant script is **TypewriterEffect.cs**, which enhances the user experience by simulating the gradual revelation of text, as if it were being typed in real-time. This visual effect contributes to the dynamism of conversations, improving the pacing of interactions by replicating the natural flow of human dialogue. The script is customizable, allowing for adjustments in text reveal speed depending on the context, whether for fast-paced or slower interactions, thereby creating a more engaging and lifelike communication experience.

These custom scripts are seamlessly integrated to provide a highly responsive and immersive user experience. By incorporating advanced external services such as OpenAI and Microsoft Azure, the project offers sophisticated features, including AI-driven conversations and real-time speech recognition, while ensuring high levels of customization and control over the user interface. Additionally, the combination of Unity's native functionalities with cloud-based APIs highlights the potential to build complex, interactive systems within a virtual reality environment. These scripts have been meticulously coordinated to maintain an intuitive interface, allowing users to interact naturally with AI through text or speech, delivering a smooth and effective learning experience.

3.5 Challenges, Issues, and Adaptations

The development of this project faced numerous technical challenges, particularly regarding the complexities of the Unity engine and the integration of external APIs. This section outlines the key challenges encountered, the learning and problem-solving strategies adopted, and the solutions that ensured successful project completion.

3.5.1 Complexity of Unity

One of the main challenges was dealing with the vast and varied ecosystem in Unity engine. The complexity of the engine, including multiple versions, asset libraries, and APIs, often resulted in conflicts that required significant troubleshooting. This became especially challenging with specific APIs such as the voice input API and Passthrough API, which required precise configurations to

work effectively in the VR environment. For example, this project was originally intended to implement the Depth API and Meta Voice SDK; however, Unity presented pipeline conflicts, such as incompatible GUIDs and non-functional components. To keep the project on track, it became necessary to abandon the Depth API and substitute the Meta Voice SDK with Microsoft Azure services to ensure progress and meet deadlines. The process of identifying and resolving these issues was time-consuming and impacted the overall project timeline, requiring adjustments to the agile methodology.

3.5.2 C# Programming Challenges

C# was the primary scripting language used throughout the project. While powerful for game development, it presented a steep learning curve compared to beginner-friendly languages such as Python and Java. C# is more robust and complex frameworks required mastering, particularly when managing asynchronous programming for real-time features such as voice input and output. Debugging and optimizing asynchronous processes proved difficult, especially when ensuring smooth and efficient real-time user interactions, requiring significant problem-solving and optimization.

3.5.3 API Integration Difficulties

Integrating external APIs, such as OpenAI for natural language processing and Microsoft Azure for speech recognition, posed additional challenges. These APIs needed to work seamlessly within Unity's real-time environment, yet communication delays and latency initially hindered smooth user interactions. Achieving real-time conversational capabilities required careful optimization of how API requests were handled and processed. Balancing Unity's native systems with external APIs required continuous adjustments to ensure stability and responsiveness.

3.5.4 Solutions and Learning Process

To tackle these challenges, an agile and systematic approach was employed, breaking down complex problems into manageable tasks. Individual APIs were thoroughly tested and debugged prior to integration into the broader system, allowing potential conflicts to be identified and addressed early on. Learning resources, such as YouTube tutorials and Unity forums, proved invaluable for resolving technical issues, offering practical insights into scripting, asset management, and API integration. Cross-referencing multiple tutorials enabled the team to pinpoint effective strategies for overcoming challenges related to API integration and script optimization.

3.5.5 Modular Design Approach

A modular design strategy was adopted to streamline development and mitigate complexity. By developing and testing individual components—such as the voice input system, AI response mechanisms, and VR interactions—independently, the team could troubleshoot and optimize each element without disrupting the overall system. This modular approach provided flexibility, allowing components to be iterated upon or replaced without necessitating major structural changes to the project.

3.5.6 Simplicity in Development

Simplicity was prioritized throughout the project. Overly complex solutions often introduced additional challenges, particularly in terms of compatibility and performance. By focusing on clear and straightforward design choices, conflicts between Unity assets, APIs, and scripts were minimized. Maintaining a simple and organized project structure helped reduce errors and facilitated the seamless integration of new features as the project evolved, ensuring the project remained on schedule.

3.5.7 Final Adaptations and Optimizations

As the project advanced, continuous testing and refinement were essential to resolve persistent issues. Regular feedback loops ensured that components such as voice recognition and AI responses operated smoothly in real time. Gradual enhancements were achieved through iterative learning, a modular design approach, and an emphasis on simplicity. These strategies allowed the project to overcome early challenges, resulting in a fully functional, responsive system capable of real-time interaction within the VR environment.

3.6 Limitations and Recommendations

While the project achieved notable successes, particularly in facilitating multi-language interactions and guiding users through meaningful conversations, several limitations impacted the overall user experience.

3.6.1 Limitations

A major success was the AI's ability to enable seamless multi-language conversations. However, the AI lacked emotional expressiveness, exhibiting limited facial expressions, gestures, and vocal variations. This absence of emotional depth made interactions feel less lifelike and engaging, especially in a VR setting where human-like interaction is essential for maintaining user interest.

The virtual environment, while functional, remained relatively static. The restricted range of gestures and the inability to interact meaningfully with objects or the environment diminished the overall sense of immersion, potentially affecting user engagement in the learning process.

Furthermore, the absence of a game-objective system reduced user motivation. Without defined goals or rewards, users found it challenging to track their progress or remain incentivized to interact consistently with the AI, which made sustaining long-term engagement more difficult.

3.6.2 Recommendations

The overall emotional expressiveness could be improved by utilizing tools that enhance facial animations and body language, such as Unity's Cinemachine and Animation Rigging. This would enable the AI to display emotions like empathy or excitement, making conversations feel more natural. Further improvements in interaction quality could be achieved by employing more advanced natural language processing models, which would allow the AI to better express emotional cues.

This would also enable smoother animations and allow the AI to perform more gestures within the environment. Using Unity's Timeline in combination with an Animator Controller would add realism and enhance the dynamic nature of the virtual world.

By incorporating tools like **Playmaker** or **GameFlow**, the project could introduce a game-objective system, where users are clearly guided through tasks and rewards, particularly in language learning. Such a system would encourage greater user engagement and offer a tangible sense of progress, making the experience more fulfilling.

In summary, while the project succeeded in providing valuable multi-language interaction, reducing limitations related to emotional expressiveness, environmental realism, and user engagement would significantly enhance both the educational value and overall immersion of the application.

3.7 Ethical Considerations

The solution was developed with a strong emphasis on ethical considerations, including user privacy, fairness, and psychological well-being. Data privacy was a primary concern. The system was designed to avoid storing personal data, conversation logs, or any related memory information, with all data stored locally. This approach complies with data protection regulations and safeguards the confidentiality of user interactions.

Bias mitigation was also a key focus, particularly in regard to Language Learning Models, which can sometimes reflect biases related to gender, culture, or race. To address this, specific prompt engineering techniques were employed to guide the AI toward generating neutral and inclusive responses. For example, by restricting the range of language use and carefully crafting prompts, the likelihood of biased AI output was minimized. This ensures that the system does not reinforce harmful stereotypes and promotes fairness in interactions, making the AI accessible and unbiased for users from diverse cultural backgrounds.

In terms of psychological well-being, risks associated with the immersive nature of VR, such as discomfort, disorientation, or potential addiction, were carefully considered. To mitigate these risks, the application includes features that encourage users to take regular breaks and limit session durations. The system is also designed to prevent manipulative behaviors and inappropriate language through prompt engineering, ensuring that the AI's responses are crafted to avoid coercive or harmful interactions. Additionally, language filters were implemented to restrict the AI from generating offensive or harmful content, thereby maintaining a positive and ethically sound environment for users.

In summary, this project prioritizes privacy, inclusivity, and user well-being, ensuring a secure, unbiased, and ethically sound user experience through techniques such as prompt engineering and careful control over AI interactions.

Chapter 4 Testing and Evaluation

4.1 Software Testing – Strategy and Evaluation

The software development process was designed to support an iterative testing approach, combining technical functionality tests with user evaluations for continuous improvement. The iterative testing process integrated both technical functionality and user evaluation. The software development followed an iterative testing methodology, where each feature was rigorously assessed in a real-world XR environment using Meta Quest 2. This approach facilitated the identification of issues related to both system performance and user experience. Core functionalities such as voice recognition, 3D interaction, and controller responsiveness were technically tested to ensure optimal performance. Each iteration included extensive testing for multilingual dialogue handling, visual tracking, and API integration. Simultaneously, user testing focused on assessing usability, intuitiveness, and engagement through structured interactions in controlled environments, followed by feedback collection.

Major technical challenges, such as incompatibility with the Meta Voice SDK, were identified and mitigated over several development cycles. The Microsoft Azure SDK was implemented, offering high accuracy in multilingual voice recognition for English, Chinese, and Japanese. Testing also uncovered latency in controller responsiveness and inconsistencies in visual tracking with the 3D model. These issues were resolved through Passthrough API optimizations and adjustments to input sensitivity. By the final stages, these refinements significantly improved interaction fluidity and system stability.

4.2 User Testing and Feedback

Testing with the Meta Quest 2 was conducted in controlled indoor environments, optimized for free movement and accurate voice recognition. Users interacted freely with features such as real-time pronunciation feedback, interactive learning scenarios, and language switching, allowing for the assessment of system effectiveness and engagement potential.

4.2.1 Participant Demographics and Language Background

The participant pool consisted of a diverse demographic, primarily individuals aged 25-30 with postgraduate qualifications. Most were native Chinese speakers learning English or Japanese, with proficiency levels ranging from beginner to fluent. This diversity provided valuable insights into the system's ability to accommodate different learner profiles. Furthermore, Participants had varying levels of experience with VR/AR/MR technologies and AI tools like ChatGPT or Siri. Many had previously used language-learning platforms such as Duolingo, offering a comparative benchmark for evaluating the XR-LLM system. Fluent English learners particularly appreciated the immersive experience provided by the system, especially the real-time feedback and interactive scenarios. Real-time pronunciation feedback and interactive learning scenarios helped create a dynamic learning environment. Participants suggested enhancing 3D avatar personalization and adjusting the virtual assistant's position to replicate a more

traditional classroom experience. While the voice command functionality was highly rated, minor delays in language switching prompted refinements to the recognition algorithms.

The system was regarded as more engaging than traditional learning methods, thanks to real-time feedback and immersive AR elements that created a personalized tutoring experience. Users reported full immersion during sessions, with AI-powered feedback enhancing language retention. Recommendations included incorporating role-playing simulations for real-world conversational practice. Some users experienced lag during AR scene transitions or when interacting in confined spaces, highlighting the need for backend optimizations. Additional suggestions included further customization options for avatars and improved subtitle synchronization to align real-time spoken dialogue more accurately.

4.3 Review of Feedback and System Improvement

Early issues with language switching were addressed by refining recognition algorithms, reducing the need for manual interventions. Interface layout challenges, such as overlapping text boxes, were resolved to improve readability. Additionally, backend optimizations reduced latency and enhanced system performance during scene transitions. The iterative technical and user-centered testing process resulted in a more reliable, user-friendly, and engaging system, with targeted improvements driven by user feedback.

4.4 Future Directions

Future development will focus on optimizing system performance, particularly by improving loading times during AR scene transitions and dialogue processing. Refining the backend architecture will ensure smoother interactions, especially during complex scenarios. Enhancing language-switching functionality will be a priority to reduce delays and enable real-time transitions as the system scales for more complex multilingual interactions. Further improvements in avatar personalization, dynamic animations, and facial expressions will increase user engagement and immersion. Additionally, improving subtitle synchronization for longer interactions is a key area of focus. Expanding learning scenarios, such as role-playing simulations, will allow learners to apply language skills in practical contexts. Future research should explore how the system can be extended to support other educational disciplines, showcasing the versatility of integrating XR technology with LLMs.

4.5 Conclusion

The iterative development process successfully refined the system, achieving the goal of providing a multilingual, interactive learning experience through XR and LLMs. Continuous feedback and real-time adjustments enhanced both technical performance and the user experience. Although the system performs well, future efforts should focus on further optimization, increased interactivity, and deeper engagement. These enhancements have the potential to significantly impact various educational environments, making learning more immersive and interactive.

Chapter 5 Conclusion

The effective integration of XR technology with LLMs for a multi-user, multimodal experience in the development of MR applications demonstrated strong performance in facilitating multilingual conversations. It enabled smooth interactions and provided flexibility to users within MR environments, successfully achieving the core project objectives.

5.1 Accomplishments and Successes

The iterative development and testing process was pivotal in refining the application's functionality and stability. The system enabled fluid conversations in Chinese, English, and Japanese, with positive user feedback on the natural flow of dialogues and the educational strategies embedded in the 3D model. The Passthrough API enhanced user interaction by enabling adjustments to the position and scale of 3D models, which significantly improved immersion. Over time, critical bugs were drastically reduced, leading to a more reliable and user-friendly system. Additionally, the switch to the Microsoft Azure SDK enhanced voice recognition, particularly in handling multilingual dialogue. The AI-generated dialogues effectively sustained conversations while offering users opportunities to practice and enhance their language skills. This approach proved to be a powerful tool for interactive language learning, especially in conversational practice.

5.2 Identified Limitations and Methodological Reflections

Several limitations were identified during the evaluation. Loading speed emerged as a significant issue, causing delays in processing complex prompts, which negatively impacted the user experience. This highlights the need for architectural optimization to improve performance. Although the language-switching functionality generally performed well, occasional failures indicated that the language context management system could benefit from further refinement to ensure seamless transitions. This is critical for maintaining user engagement across multiple languages. The lack of facial expressions and minimal movement in the 3D avatar reduced overall user engagement. Incorporating dynamic animations and facial expressions would enhance the lifelike quality of interactions, increasing the system's educational value.

The iterative testing approach effectively provided real-time feedback, enabling continuous refinements. However, the absence of formal performance metrics—such as bug tracking or benchmarking—limited the ability to assess performance under varied conditions. Future projects would benefit from incorporating these metrics for more comprehensive evaluations.

5.3 Conclusion

This project successfully demonstrated the potential of integrating XR and LLMs to create a multilingual, interactive learning tool. While the core objectives were met, areas for improvement include performance optimization, language-

switching reliability, and enhancing the interactivity of the 3D model. Addressing these limitations will be crucial for future iterations, expanding the system's capabilities across additional educational contexts and improving its adaptability for broader multilingual applications.

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Appendix A : Prioritization Document for Love & Language XR Game Development

This appendix outlines the prioritization of tasks for the "Love & Language" XR game development project. Tasks are categorized into four priority levels: Must Have, Should Have, Could Have, and Won't Have. Each task is assigned story points to estimate the effort required, helping manage time and development efforts effectively.

- **Prioritization Categories:**
 - **Must Have:** Critical tasks that are essential for the basic functionality of the game.
 - **Should Have:** Important tasks that enhance the game but are not critical for basic functionality.
 - **Could Have:** Nice-to-have tasks that add value but can be omitted if time does not permit.
 - **Won't Have:** Tasks that are out of scope for this project.
 - **Time Units:**
 - **Sprint:** 1 Sprint = 1 Week = 10 Story Points (SP)
 - **Story Points:** 1 SP = 4 hours of work
 - **Must Have:**
1. **Unity Project Setup (1 Story Point)**
 - Create a new project in Unity.
 - Import the AR Integration package and other necessary plugins.
 2. **Character Modeling and AR APIs Setup (2 Story Points)**
 - Create basic 2D models for the AI assistant and dialogue boxes.
 - Integrate the models into the Unity project.
 3. **OpenAI API Integration (3 Story Points)**
 - Obtain and configure the OpenAI API key.
 - Write scripts to call the OpenAI API for dialogue generation.
 4. **Google Cloud API Integration (4 Story Points)**
 - Enable and configure Google Cloud Speech-to-Text and Text-to-Speech APIs.
 - Write scripts to implement real-time speech recognition and synthesis.
 5. **AI Assistant Interaction (10 Story Points)**
 - Implement basic interactions with the AI assistant.
 - Ensure the AI assistant can respond to player questions using generated dialogues.
 6. **Basic Dialogue Mechanics (10 Story Points)**
 - Set up the task system for language dialogue practice, translation challenges, and cultural knowledge quizzes.
 - Integrate the learning value system for tracking progress.
 7. **UI/UX Design (10 Story Points)**
 - Design and implement a user-friendly interface for the game.
 - Ensure clear display of learning values and task progress.
 8. **Scene Detection (5 Story Points)**
 - Implement XR scene detection to capture user interactions and detect clicked objects.

- Write scripts to recognize user actions and provide appropriate feedback.
- 9. **Learning Progression System (8 Story Points)**
 - Create a multi-stage learning progression system.
 - Adjust difficulty and content based on player's learning value.
- **Should Have:**
 1. **Sound Effects and Background Music (1 Story Point)**
 - Add sound effects and background music to enhance immersion.
 - Ensure synchronization of sound with user interactions and AI responses.
 2. **Dynamic Prompt Generation (4 Story Points)**
 - Write scripts to generate dynamic prompts based on scene and learning value changes.
 - Optimize dialogue generation for coherence and relevance.
 3. **Real-World Interaction (6 Story Points)**
 - Integrate interaction mechanisms with real-world objects and scenes.
 - Test and optimize real-world interaction features.
 4. **Initial Testing and Refinement (10 Story Points)**
 - Conduct user testing to gather feedback.
 - Refine gameplay mechanics, UI/UX, and AI interactions based on feedback.
- **Could Have:**
 1. **Advanced AI Models (5 Story Points)**
 - Integrate advanced RAG search and LoRA model training for the AI assistant.
 - Improve AI response accuracy and relevance.
 2. **Interactive 3D Models (5 Story Points)**
 - Develop a 3D model of the AI assistant that can walk around the XR scene and be interacted with.
 3. **Diverse AI Assistants (15 Story Points)**
 - Implement different personalities and genders for AI assistants.
 4. **Complete Storyline and Branch Options (10 Story Points)**
 - Design and implement a complete storyline with branching options.
 5. **Multiple Platform Version (10 Story Points)**
 - Develop a VR version for a more immersive experience.
 6. **Alternative Learning Methods (15 Story Points)**
 - Integration of other learning methods such as vocabulary practices and quizzes.
- **Won't Have:**
 1. **Complex AI Behaviors (N/A)**
 - Development of complex AI behaviors beyond basic interactive dialogue.
 2. **Large-Scale User Testing (N/A)**
 - Large-scale user testing involving hundreds of participants.
 3. **High-Resolution 3D Models and Animations (N/A)**
 - Creation of high-resolution 3D models and complex animations.
 4. **Multiplayer Modes (N/A)**
 - Implementation of multiplayer modes.
 5. **Detailed Analytics and Reporting (N/A)**
 - Integration of detailed analytics and comprehensive reporting tools.